

NON-INSECT HEXAPODS

Class Collembola

Springtails

Although small and easily overlooked, springtails are present in vast numbers in grassland, leaf litter, and soil. On the underside of the abdomen, they have a ventral tube that allows them to regulate water and grip smooth surfaces, and a fork-shaped jumping organ (furcula), which can be released like a spring to flick them away from predators. There are 32 families (about 8,100 species) of springtails.

Family Poduridae

Water springtail

Occurrence 4 species in the Northern Hemisphere; on the surface of freshwater ditches, ponds, canals, bogs



The most common and widespread species of this family, *Podura aquatica*, is up to 1/16 in (2 mm) long. It varies from brown or reddish brown to dark blue or almost black. The furcula is quite flat and long, and reaches the abdominal ventral tube, enabling the springtail to cling to the water's surface and jump effectively. A scavenger, it feeds on small, floating particles of food. Its eggs are laid among vegetation in or around water bodies.



PODURA AQUATICA
This abundant springtail is well adapted to life on water. It may gather in such numbers on freshwater surfaces that the water appears dark.

Family Sminthuridae

Globular springtails

Occurrence 245 spp. worldwide; in leaf litter, fungal fruiting bodies on water surfaces, in caves



Also known as garden springtails, these hexapods are 1/32 – 1/8 in (1 – 3 mm) in length. Ranging from pale to dark brown or green, they have almost spherical bodies. The long, elbowed male antennae in many species are designed to hold the female during mating. Eggs are laid in small batches in soil, and development to sexual maturity may take as little as one month. Several species of these herbivores, such as *Sminthurus viridis*, are pests of crop seedlings.



SMINTHURIDES AQUATICUS
This tiny hexapod—only 1/16 in (2 mm) long—is common on the surface of ponds, but does not flock in such large numbers as the water springtail.

Class Protura

Proturans

Like springtails, proturans are sometimes found in leaf litter, but most species live in the soil. These often microscopic animals do not have eyes, tails (cerci), or antennae, and instead they use their front legs as sensory organs. Their mouthparts, which can both pierce and suck, are kept in a pouch and are pushed out during feeding. There are 7 families of proturans, containing about 760 species.

Family Eosentomidae

Eosentomids

Occurrence 340 spp. worldwide; in soil, leaf litter, mosses, humus, decaying wood



These pale and soft-bodied hexapods are 1/64 – 1/16 in (0.5 – 2 mm) long, and have a conical head and an elongated body. They eat dead organic matter

and fungi. The eggs, laid in soil or leaf litter, are spherical and have patterns or warts on them. Larvae are identical to adults, but smaller.



EOSENTOMON DELICATUM
This often abundant species is native to Europe, although the genus is found all over the world. It lives in soil, especially chalky soil.

Class Diplura

Diplurans

These blind, elongated, and soft-bodied hexapods are sometimes called two-tailed bristletails due to their 2 terminal cerci. They have a large, distinct head and biting, eversible mouthparts. There are 8 families and about 975 species.

Family Japygidae

Japygids

Occurrence 408 spp. worldwide; in soil



These species are slender-bodied and pale, ranging in length from 1/4 – 1 1/4 in (0.6 – 3 cm). The end of the abdomen carries a pair of distinctive, tough,

forcepslike cerci which are used to catch small arthropod prey. Eggs are laid in batches in soil.

HOLJAPYX DIVERSIUNGUIS
Also called the slender dipluran, this common North American species has stout, terminal cerci. Its abdominal segments are pale around the edges.



INSECTS

Order Archaeognatha

Bristletails

Bristletails are wingless insects that look hump backed in side view. They have simple mouthparts, large eyes that touch each other, and 3 tails (cerci), the middle being the longest. Like silverfish, they do not undergo metamorphosis. There are about 470 species in 2 families.

Family Machilidae

Jumping bristletails

Occurrence 333 spp. worldwide; under stones, in leaf litter, decaying vegetation



These species are elongated and covered with patterns of drab brown or dark gray scales. Up to 3/8 in (1.2 cm) long, many can run and jump. They feed

on algae, lichen, and plant debris. Eggs are laid in small batches in crevices, and the young mature in 2 years. They may live up to 4 years.



PETROBIUS MARITIMUS
This fast-running insect lives on rocky shorelines in the Northern Hemisphere, often in large numbers.

Order Thysanura

Silverfish

Distinguished from bristletails by their widely separated eyes, and 3 equally sized tails (cerci), silverfish have elongated, flattened bodies, often covered in silvery scales. There are about 570 species in 6 families; some kinds are a familiar sight in houses.

Family Lepismatidae

Lepismatids

Occurrence 295 spp. worldwide, especially in warmer regions; in tree canopies, caves, human dwellings



Lepismatids are brownish, nocturnal insects with compound eyes but no ocelli. They are 3/16 – 3/4 in (0.8 – 2 cm) long. Courtship is simple, and

fertilization is indirect. Eggs are laid in small batches in cracks or crevices. Domestic species feed on flour, damp textiles, wallpaper paste, and book bindings. Lepismatids may live up to 4 years.



LEPISMA SACCHARINA
The common silverfish emerges only at night, prefers damp places, and can be found in kitchens, bathrooms, and basements worldwide.